Practice problems #13

1. Find the inverse of the matrix below, if exists.

$$\begin{bmatrix} 1 & 0 & 5 \\ 1 & 1 & 0 \\ 3 & 2 & 6 \end{bmatrix}$$

2. True or False. Justify.

   (a) Suppose A and B are $n \times n$, B is invertible, and AB is invertible, then A is invertible.

   (b) It is possible for a 6×6 matrix Q to be invertible when its columns do not span $\mathbf{R^6}$.

   (c) Suppose $AB = AC$, where B and C are $n \times p$ matrices and A is invertible. Then $B = C$.

   (d) If A is invertible, then the columns of A^{-1} form a linearly independent set.

   (e) If A and B are both square matrices and AB is invertible, then A is invertible.

Answers or Hints are on page 2.

#1:

$$\begin{bmatrix} 6 & 10 & -5 \\ -6 & -9 & 5 \\ -1 & -2 & 1 \end{bmatrix}$$

#2(a): True. Let $C = AB$, and get the equation $A = CB^{-1}$. From the given hypothesis, C is invertible, and B^{-1} is invertible (why?). A product of invertible matrices is ?
Caution: you can't use A^{-1} anywhere in the proof because you don't know if it exists until you finish the proof!
#2(b): False. Justify using IMT.
#2(c): True. why?
#2(d): True. If A is invertible, is A^{-1} invertible?
#2(e): True. What is the size of AB? Is there a square matrix C so that $(AB)C = I$? What is the size of BC? (Note that you *cannot* assume that A and B separately is invertible!)